Custom script

Version 1.1.0, by Giorgio Bianchini

Description: Executes custom code.

Module type: FurtherTransformation

Module ID: `a76d00d2-95e0-4274-a77d-1439a013e3d9`

This module makes it possible to execute custom C# code to transform the tree. This can be useful either to perform one-off complicated modifications of the tree, or as a first step in developing a new module for TreeViewer.

Parameters

Description

Control type: Text box

Default value: Describe the script

This parameter can be used to provide a short description to quickly identify what the module does without having to look at the source code. It is ignored by the module.

Source code

Control type: Source code

Default value:

```
using PhyloTree;
using System.Collections.Generic;
using TreeViewer;
using System;

namespace a6872ecb8c03346ec905f3dfbe99de00e
{
    //Do not change class name
    public static class CustomCode
    {
        //Do not change method signature
        public static void PerformAction(ref TreeNode tree,
 TreeCollection trees, InstanceStateData stateData, Action<
double> progressAction)
```

```
        {
            //TODO: do something with the tree
        }
    }
}
```

This parameter contains the source code of the script. The arguments to the `PerformAction` method are as follows:

- `tree`: the transformed tree that has been computed by the Transformed module and any preceding Further transformation modules.
- `trees`: the collection of trees that were originally read from the file.
- `stateData`: an `InstanceStateData` object that can be used to access features in way that does not depend on the program running in command-line or GUI mode.
- `progressAction`: as your script does its thing, it should invoke this `Action` with a value between 0 and 1 to give feedback to the user about the progress.

Further information

The difference between this module and the other module with the same name is that this module acts as a Further transformation, while the other *Custom script* module (id `cdb74bfb-8a90-48b3-815a-8f908d2a1ff5`) is instead a Plot action.

The code in the module can do anything, including loading additional data from a file on disk. However, this is discouraged, becaus it ties the tree file on the computer it was created on. A better approach to load additional data would be to import the data file as an attachment and read the data from the attachment. Attachments can be accessed using the `Attachments` property of the `stateData` object that is passed as a method parameter.

Furthermore, since the code in the module can do anything, it may also be a security risk to open files originating from unknown sources; thus, you should either make sure that any file you open comes from a reputable source, or avoid loading source code from tree files at all.